

BEI 400-500

FORWARD VIEW
Distance 0
1-2-3-4-5
6-7-8-9

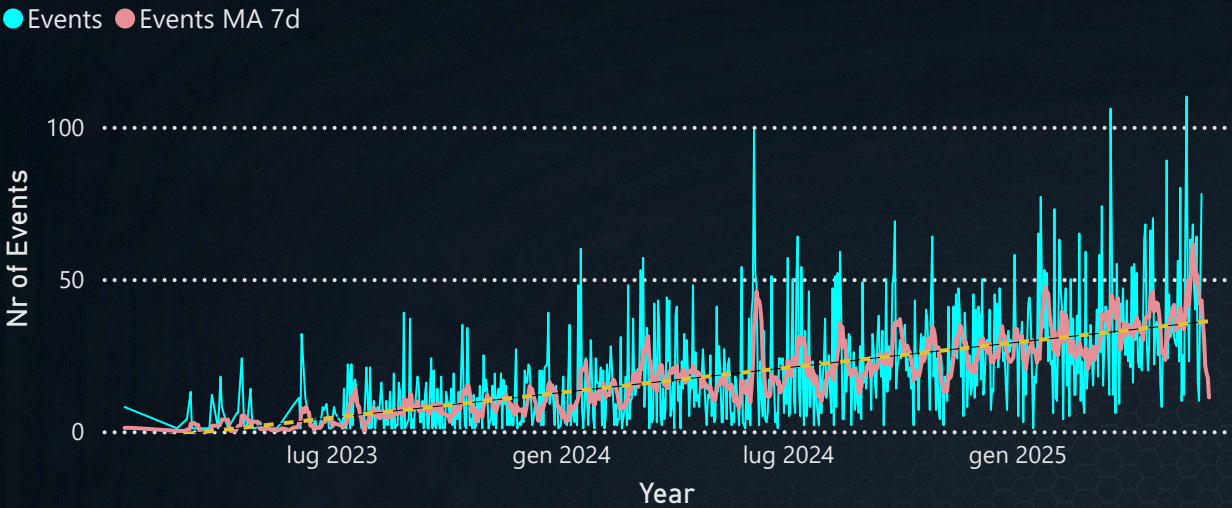
NEXUS: Facility Brain

Predictive Maintenance explorer

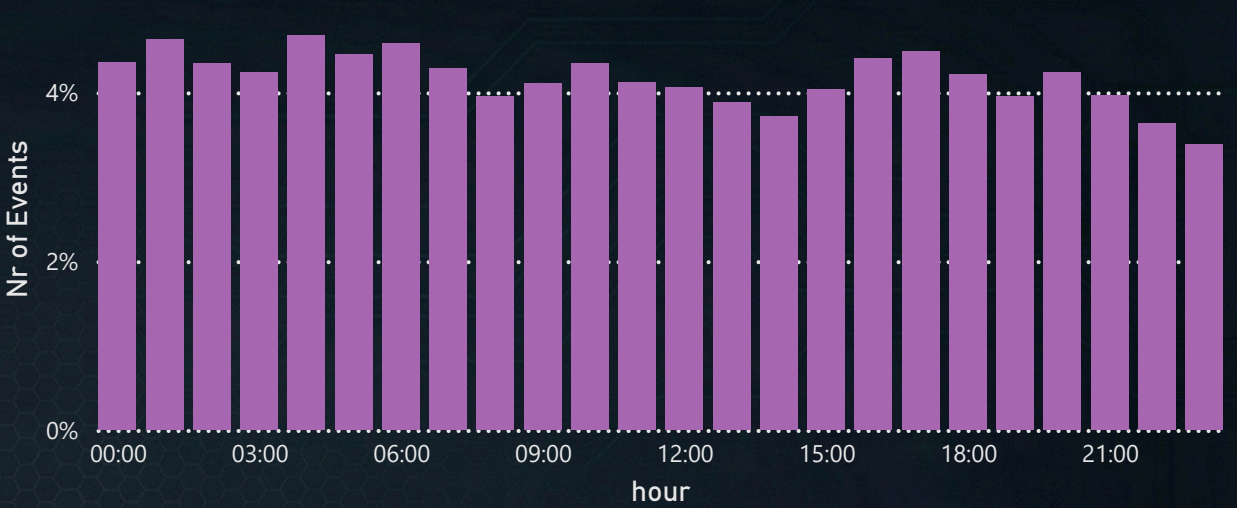


Maintenance events during the period of observation

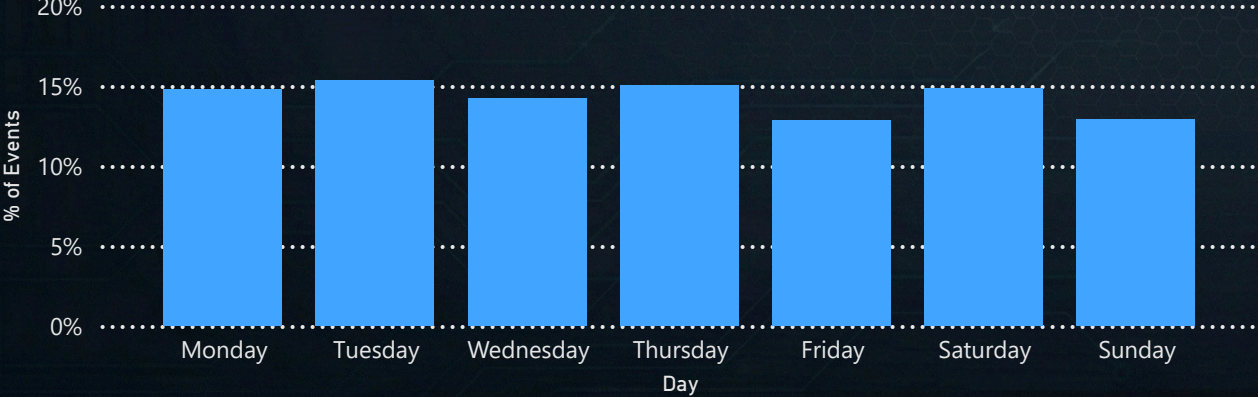
Nr. of Events and MovingAV 7gg per period



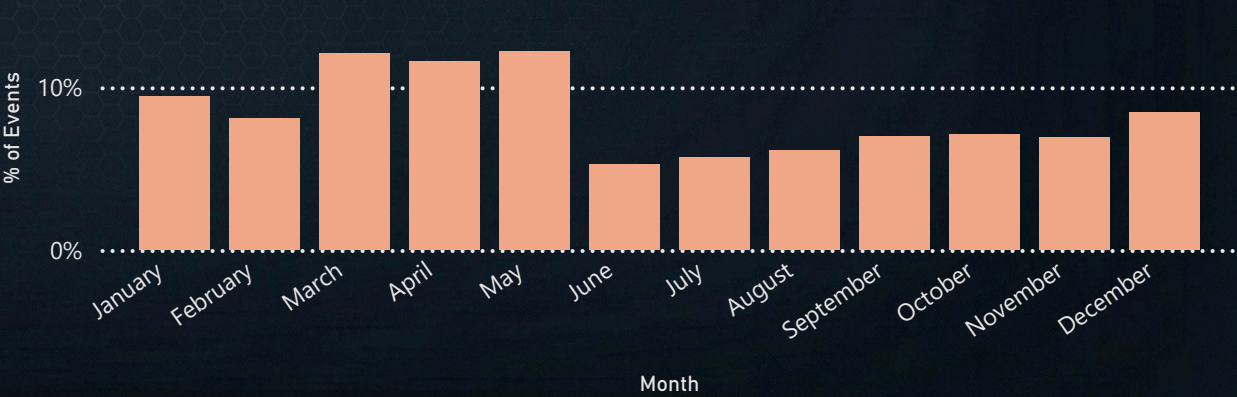
Hourly distribution of Maintenance events



Maintenance Events distribution by day of the week

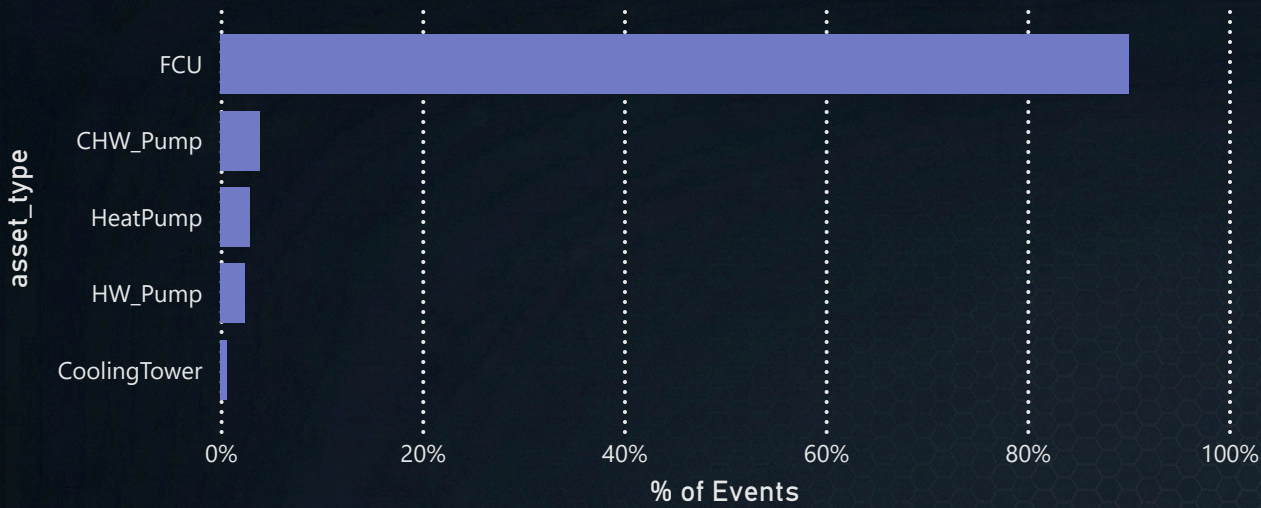


Maintenance Events distribution by month

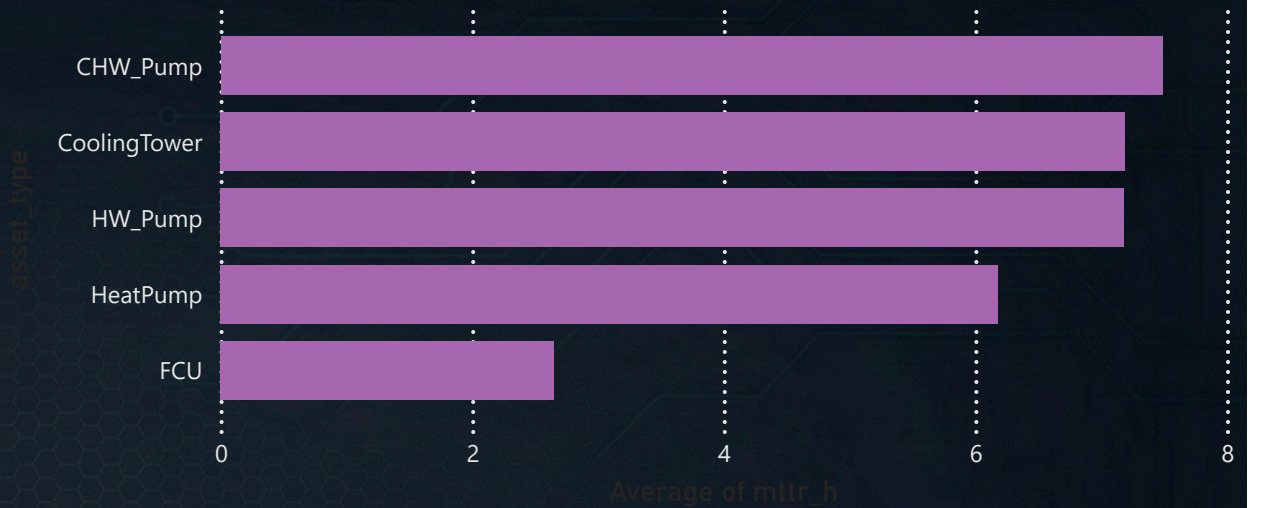


Characteristics of maintenance events

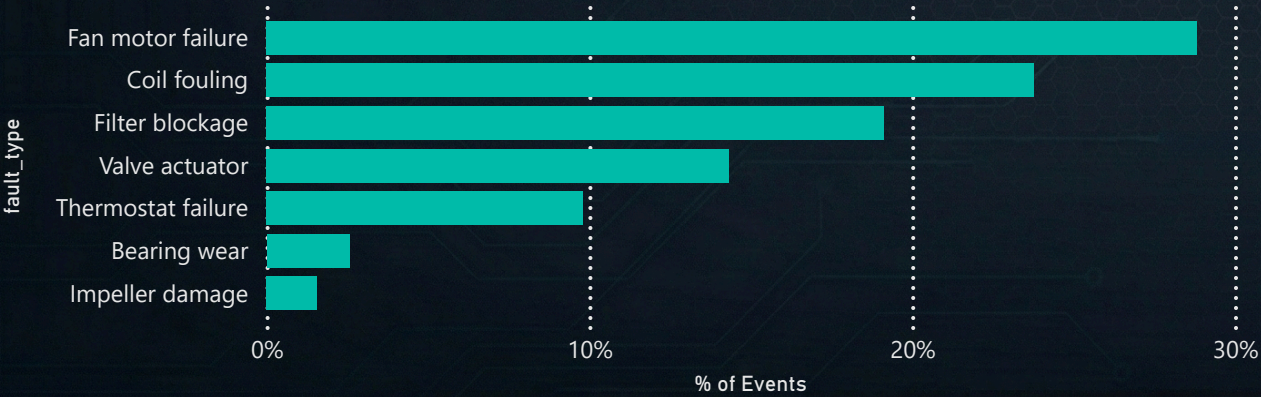
Maintenance Events by Asset Type



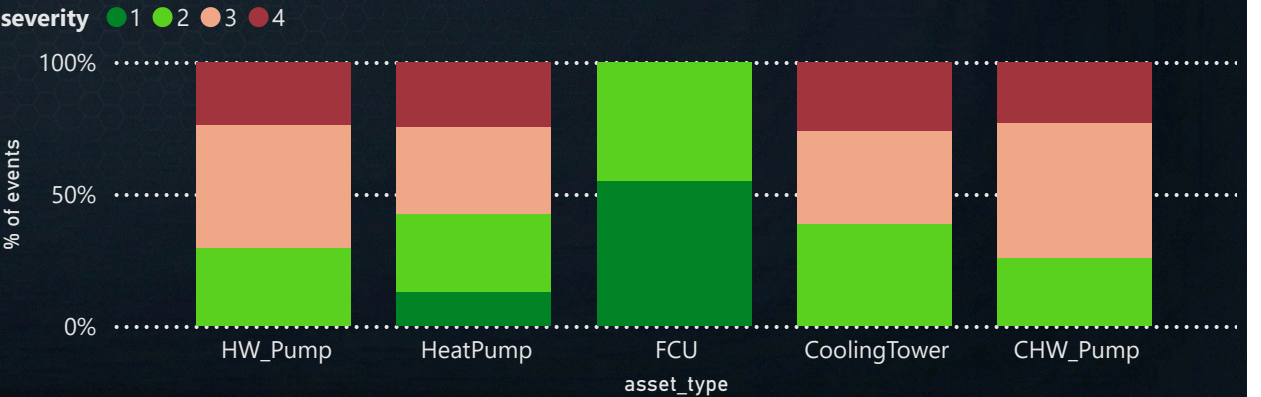
Mean MTTR by Asset Type



TOP 7 Faults type

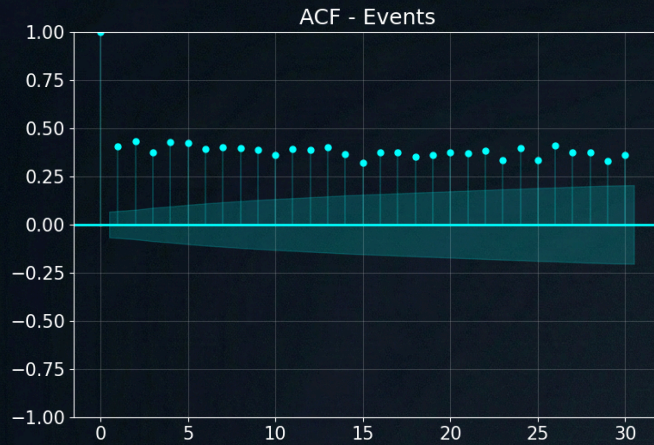


Maintenance Events distribution by asset type and severity

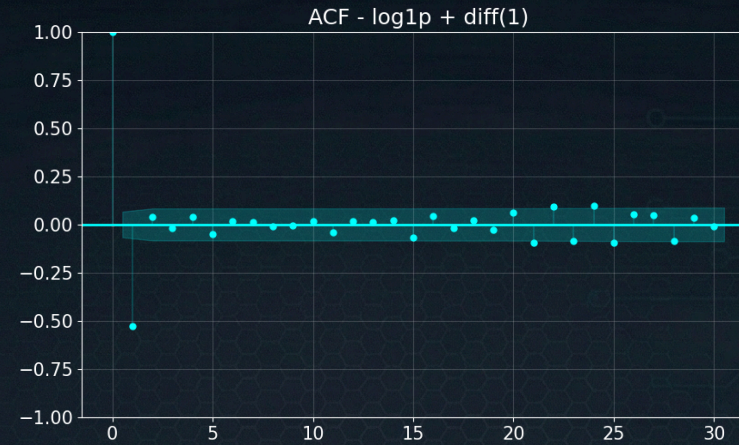


Time series analysis of maintenance events

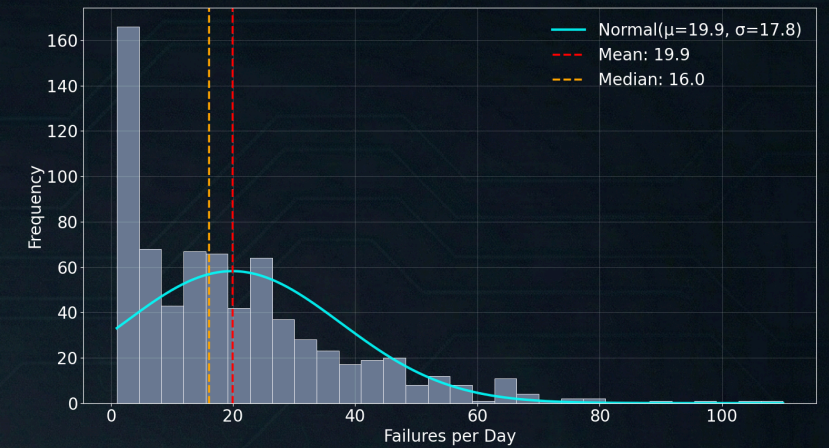
Time Series Analysis - ACF



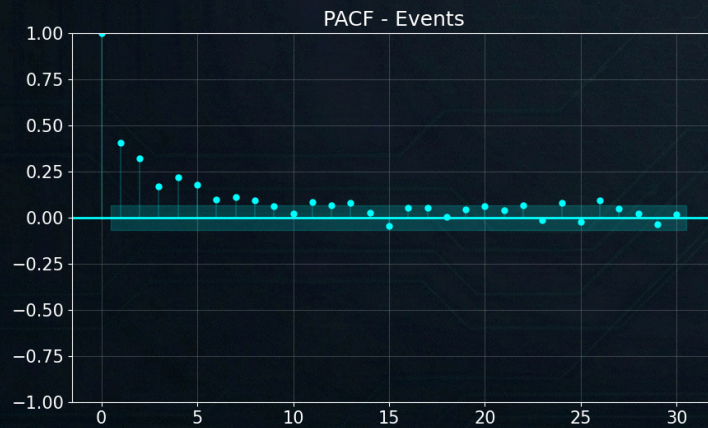
Time Series Analysis: Stationarity Inspection - ACF



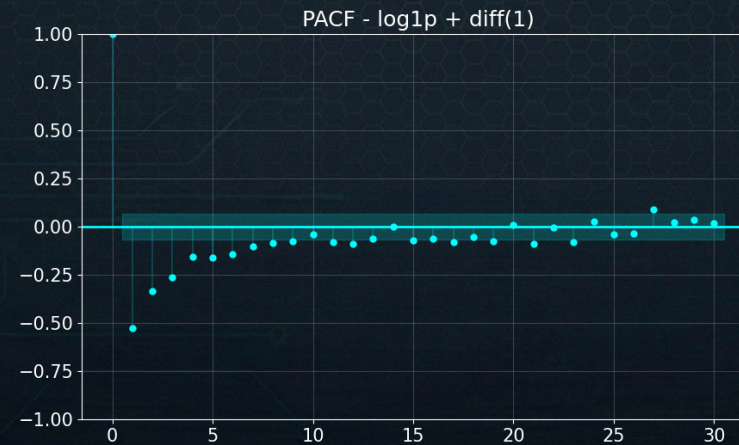
Distribution of Daily Failure Counts



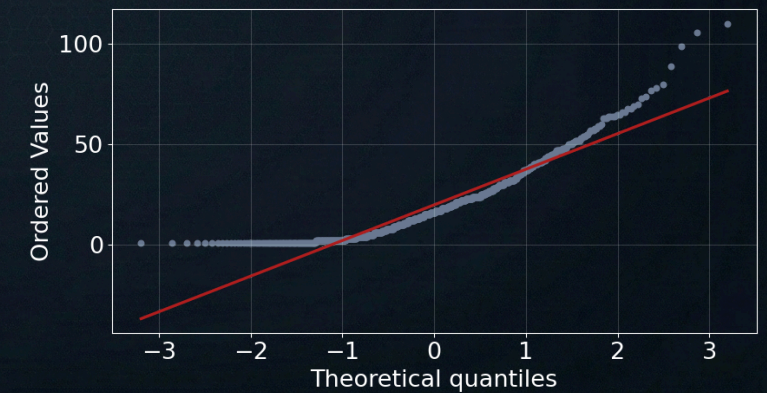
Time Series Analysis - PACF



Time Series Analysis: Stationarity Inspection - PACF

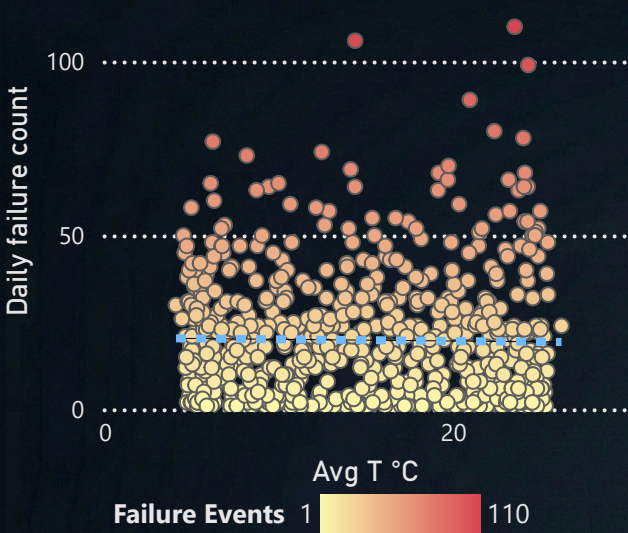


Q-Q Plot (Normality Test)\nNon-Normal Distribution Confirmed

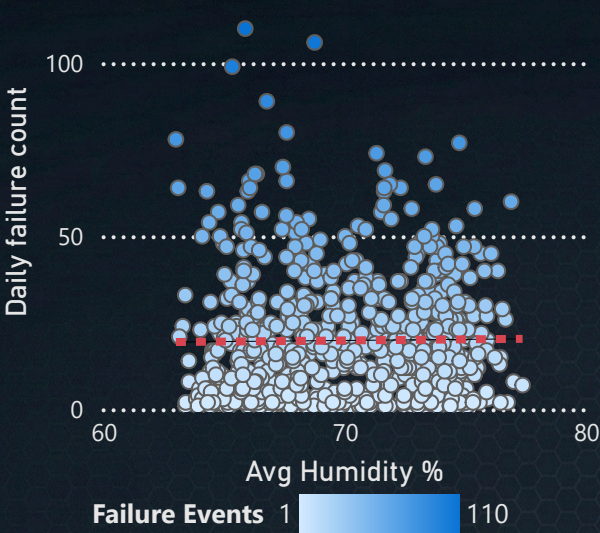


Maintenance events: Correlations

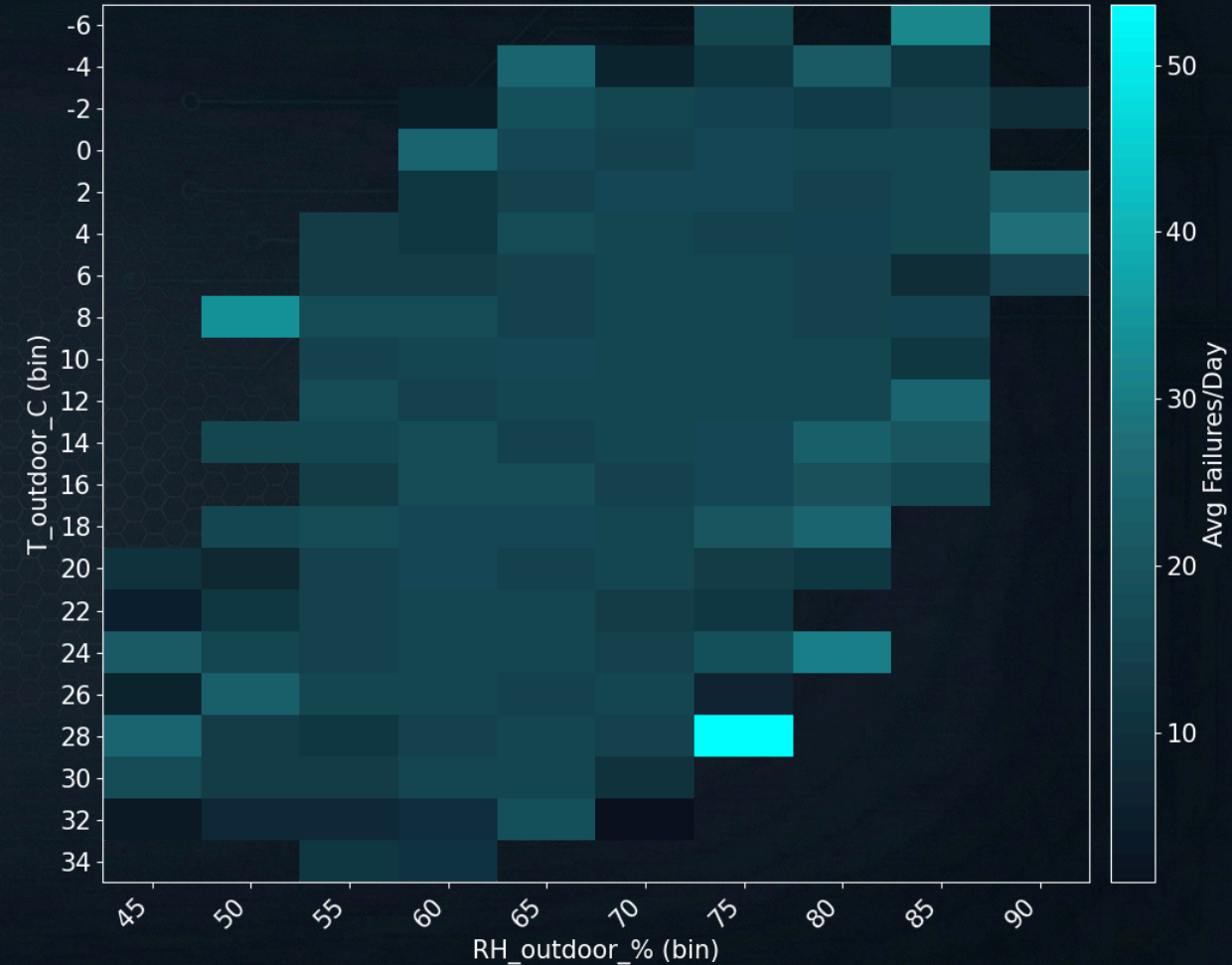
Failures vs Temperature (°C)



Failures vs Humidity (%)



Avg Failures/Day Heatmap — Temperature * Humidity Interaction



Mean Time To Repair, Downtime and Inventory impact Analysis

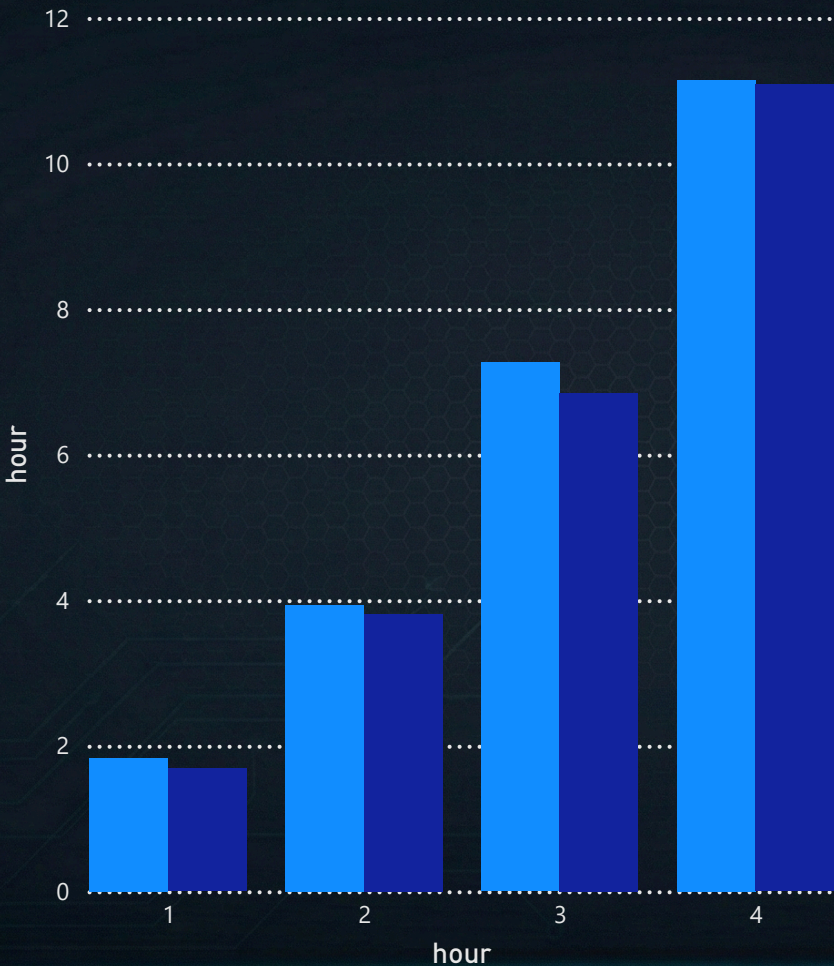
Select Severity of Mainenance Event

severity

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

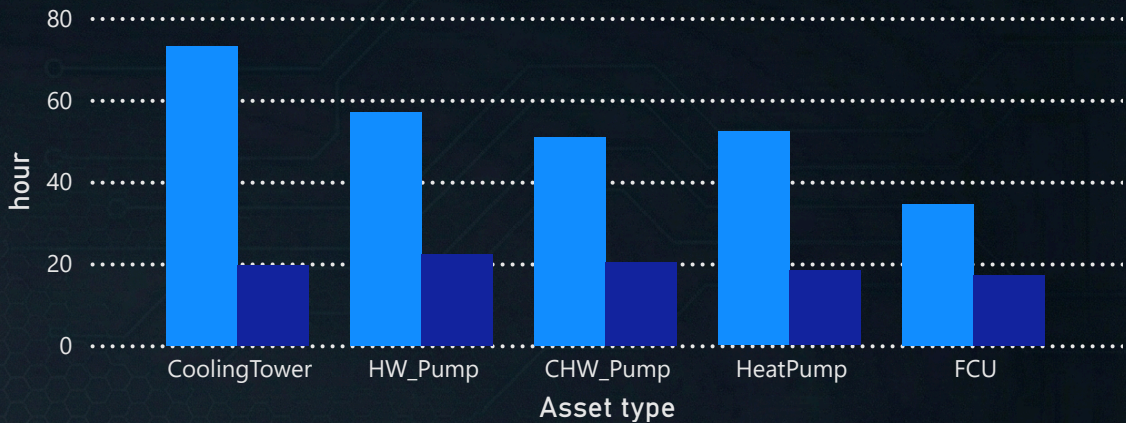
Mean time to repair by severity and stock status

part_in_stock ● 0 ● 1



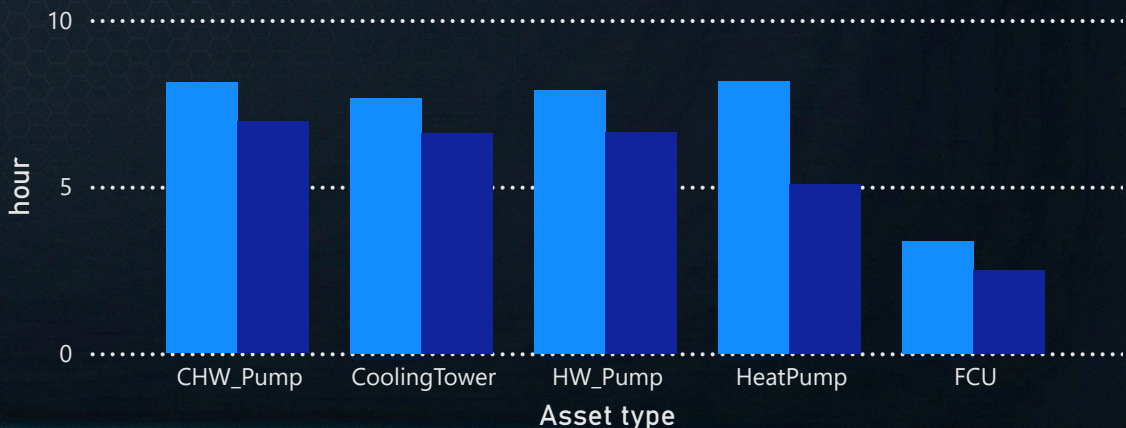
Mean Downtime by asset type and stock status

part_in_stock ● 0 ● 1



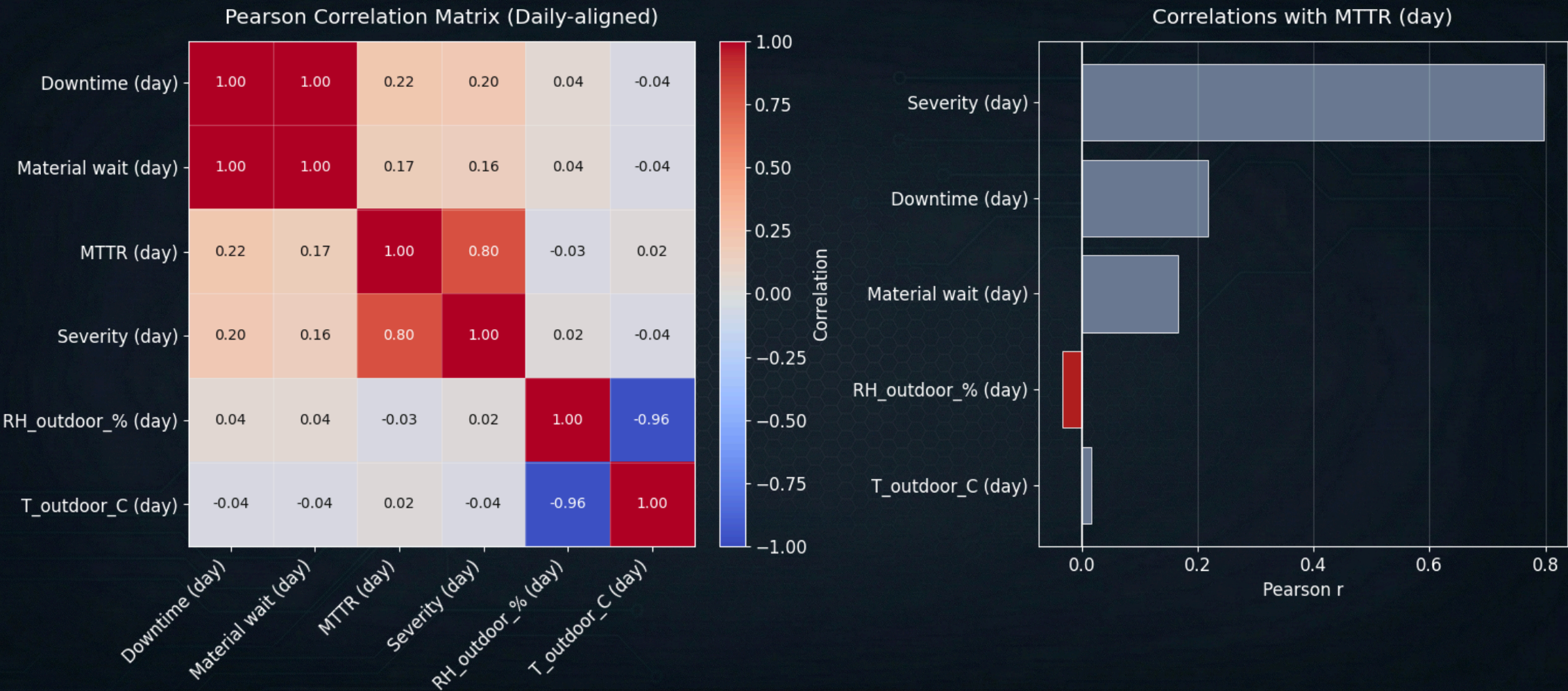
Mean time to repair by asset type and stock status

part_in_stock ● 0 ● 1



MTTR root cause insights: operational and enviromental correlation analysis

Pearson Correlations: MTTR, Operational Drivers, and Weather (Daily Aggregation)



Interactive predictive maintenance explorer

Select Severity of Maintenance Event Select in stock material

severity

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

part_in_stock

- ☐ 0
- ☐ 1

Select Period

Rolling_date

01/11/2022

31/12/2025

Select Asset type

asset_type

- ☐ CHW_Pump
- ☐ CoolingTower
- ☐ FCU
- ☐ HeatPump
- ☐ HW_Pump

Nr. of Events and MovingAV 7gg per period

● Events ● Events MA 7d

